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LAB SESSION 3:CLASS DIAGRAM

→ List of Classes:

1. User
2. Student
3. Librarian
4. Admin
5. Notification
6. Borrow
7. Reservation
8. Fine
9. Book

→ Class Responsibilities (CRC Approach):

- **Class: User**
  - **Attributes:** userId, name, email, status
  - **Methods:** searchBook(), viewBorrowHistory(), preBook(), cancelReservation(), getNotifications()
- **Class: Student**
  - **Attributes:** studentId
  - **Methods:** viewMonthlyReport()
- **Class: Librarian**
  - **Attributes:** librarianId
  - **Methods:** addBook(), removeBook()
- **Class: Admin**
  - **Attributes:** adminId
  - **Methods:** createAccount(), deleteAccount(), blockAccount()
- **Class: Book**
  - **Attributes:** bookId, title, author, language, totalCopies, availableCopies
  - **Methods:** checkAvailability()
- **Class: Borrow**
  - **Attributes:** borrowId, bookId, userId, issueDate, returnDate
- **Class: Reservation**
  - **Attributes:** reservationId, bookId, userId, reservationDate, status
- **Class: Fine**
  - **Attributes:** fineId, userId, borrowId, amount, fineDate, isPaid
  - **Methods:** calculateFine()

- **Class: Notification**
    - **Attributes:** notificationId, notificationType, message, userId, dateTime

→ Class Diagram Output:

- **Inheritance:** Student, Librarian and Admin inherits User.
- **Association:**
    - Admin — User: manages user accounts
    - User — Borrow: creates borrowing records
    - User — Reservation: creates/cancels reservations
    - User — Notification: receives notifications
    - Librarian — Book: manages books
    - Borrow — Fine: generates fine
    - Reservation — Book: reserves book
    - Borrow — Book: records borrowed book
    - Fine — Book: identifies associated book
- **Composition:**
    - Borrow — Book: includes the specific book being borrowed

